

OOPS ASSIGNMENT

Question1.

Code :

```
#include <iostream>
```

```
using namespace std;
```

```
class Book {
```

```
public:
```

```
string title, author, ISBN;
```

```
}; class Library {
```

```
Book books[10]; int count;
```

```
public:
```

```
Library() { count = 0; } bool addNewBook(string &title, string &author, string &ISBN) { if (count >= 10)
```

```
return false; books[count].title = title; books[count].author = author; books[count].ISBN = ISBN;
```

```
count++; return true;
```

```
}
```

```
bool removeBooks(string &ISBN); void displayDetails() { for (int i = 0; i < count; i++) { cout <<
```

```
books[i].title << " " << books[i].author << " "
```

```
<< books[i].ISBN << endl;
```

```
}
```

```
}
```

```
};
```

```
bool Library::removeBooks(string &ISBN) { for (int i = 0; i < count; i++) { if (books[i].ISBN == ISBN) { for
```

```
(int j = i; j < count - 1; j++) { books[j] = books[j + 1];
```

```
} count--;
```

```
return true;
```

```
}
```

```
}
```

```
return false;
```

```
} int main()

{ Library lib; string t, a, i;

t="A"; a="AA"; i="1"; lib.addNewBook(t,a,i);

t="B"; a="BB"; i="2"; lib.addNewBook(t,a,i);

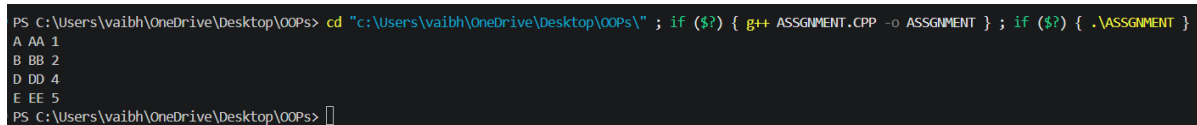

t="C"; a="CC"; i="3"; lib.addNewBook(t,a,i);

t="D"; a="DD"; i="4"; lib.addNewBook(t,a,i);

t="E"; a="EE"; i="5"; lib.addNewBook(t,a,i); string rem = "3"; lib.removeBooks(rem);
lib.displayDetails();

}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
A AA 1
B BB 2
D DD 4
E EE 5
C CC 3
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> |
```

Question2.

```
#include <iostream>

using namespace std;

class Book {

public:

string title, author, ISBN;

// default Book() {}

// parameterized

Book(string title, string author, string ISBN)

{ this->title = title; this->author = author; this->ISBN = ISBN;

}

// copy

Book(const Book &b) {

    this->title = b.title;

    this->author = b.author; this->ISBN

= b.ISBN;

} void display() { cout << title << " " << author << " "
```

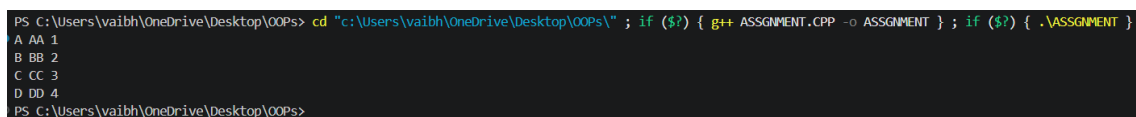
```
<< ISBN << endl; }

};

int main() {
// initializer list
Book arr[2] = {
    Book("A", "AA", "1"),
    Book("B", "BB", "2") };
for (int i = 0; i < 2; i++)
    arr[i].display();
// dynamic array
Book *b = new Book[2]{
    Book("C", "CC", "3"),
    Book("D", "DD", "4") };
for (int i = 0; i < 2; i++)
    b[i].display(); delete[] b;

}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "C:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
A AA 1
B BB 2
C CC 3
D DD 4
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question3.

Code:

```
#include <iostream>

using namespace std;

class Account {
    const long account_number; long transaction_id; string transaction_type; double balance;
public:
    Account(long acc, double bal) : account_number(acc) { balance = bal; transaction_id = 0;
}
```

```

long depositAmount(const long to, const long from, const double amount) { balance += amount;
transaction_type = "credit"; return
++transaction_id;
} long creditAmount(const long to, const long from, const double amount) { balance -= amount;
transaction_type = "debit"; return
++transaction_id;
} void displayDetails() const { cout
<< "Acc: " << account_number << " Balance: " << balance

<< " Last: " << transaction_type << endl;
} }; int
main() {
Account a1(1,1000), a2(2,2000), a3(3,3000), a4(4,4000), a5(5,5000);
a1.depositAmount(1,2,500); a2.creditAmount(2,1,300); a1.displayDetails(); a2.displayDetails();
a3.displayDetails(); a4.displayDetails(); a5.displayDetails();
}

```

Output:

```

PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Acc: 1 Balance: 1500 Last: credit
Acc: 2 Balance: 1700 Last: debit
Acc: 3 Balance: 3000 Last:
Acc: 4 Balance: 4000 Last:
Acc: 5 Balance: 5000 Last:
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>

```

Question4.

Code:

```

#include <iostream>

using namespace std;

class B;

class A { int
x; public:

A() { x = 10; }

friend int sum(A, B);

```

```
}; class B { int y; public: B() { y = 20;
} friend int sum(A, B);
}; int sum(A a, B b) { return a.x + b.y; } int main()
{
A a; B b; cout << "Sum = " << sum(a, b); }
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "C:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Sum = 30
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question5.

Code:

```
#include <iostream>
using namespace std;
class Complex {
    float real, imag;
public:
    Complex(float r = 0, float i = 0)
    { real = r; imag = i;
    }
    Complex(const Complex &c) { real = c.real;
    imag = c.imag;
    } void display() { cout << real << " + "
    << imag << "i\n"; } friend Complex sum(Complex, Complex);
};
Complex sum(Complex c1, Complex c2)
{ Complex temp; temp.real = c1.real + c2.real; temp.imag = c1.imag + c2.imag; return temp;
} int main() { Complex c1(2,3),
c2(4,5);
Complex c3 = sum(c1,
c2);
c3.display(
);
```

```
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
6 + 8i
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```